

An Environmental Simulating Chamber (ESC)

Controllable FSO Variable

- Fog
- Airspeed
- Temperature

Components/ **supplier part number / manufacturer/ Mouser/DGT/website/**

Component	Price (\$NZD)	Quantity	Total (\$NZD)	Noted
Raspberry Pi5	NA	1	0	Controller
Camera: AI Camera 12.3 MP Sony IMX500 Intelligent Vision Sensor RP2040	NA	1	0	Fog measure
Raspberry Pi 360° Rotation Portable & Mini Tripod for Raspberry Pi HQ Camera (SEVRBP0263) or Mobile Phone (55-80 mm Width) /w Phone Clamp	17.16	1	17.16	Camera stand
Temperature sensors: DHT22	10	3*	0	Temperature sensors
Pressure and Humidity sensors: BMP280	NA	2*	0	Humidity sensor
Temperature controllers: Peltier module TEC1-12706 & Heatsink	8	6*	0	Surface Heater & Cooler
High-performance thermal glue	9.23	1*	0	Thermal glue – heat sink and fan
DC Fan	3.07	6	18.42	Heater & Cooler fan
Air controllers: PWM Fan	6.16	4	24.64	Wind controller
Mist controllers:	9	2*	0	Mist controller
LED	NA	3	0	Status PWR LED
DuinoTech Arduino Compatible W2812B RGB LED Strip 2m	46.90	1	46.90	LED Stripe www.jaycar.co.nz
RGB LEDs: KY-016	NA	1	0	Status LED
Adapter 3.3V 5W	14.44	1	14.44	3.3V DC source - sensor
Adapter 5 V 15W	18.81	1	18.81	5V DC source - Fan
Adapter 12V 50W	20.28	1	20.28	12V DC source - Peltier
Solid State Relay - 5VDC Control AC Load	12	6	48	Relay for Peltier
GPIO Ribbon cable & T-Cobbler Plus for Raspberry Pi	14.90	1	14.90	Cable
PCB Board: PC Boards Vero Type Strip - 95 x 152mm	11.90	3	35.7	Board
Chamber: Plaskolite Acrylic Sheet 762 X 1524 X 2.5mm Clear	159	1	159	https://www.mitre10.co.nz/shop/plaskolite-acrylic-sheet-762-x-1524-x-2-5mm-clear/p/270040
			418.25	

Software

Python 3.13.5
 Tkinter Tk 8.6 / Tcl 8.6
 Matplotlib 3.10.1
 Pandas 2.2.3
 Numpy 2.2.4
 Pillow 11.1.0
 opencv 4.10.0
 requests 2.32.3
 gpiozero 2.0.1
 Colorzero 2.0
 adafruit-circuitpython-dht 4.0.10
 adafruit-circuitpython-bmp280 3.3.9

adafruit-blinka 8.67.0
picamera2 0.3.33
openpyxl 3.1.5
math x.x.x

Design

- Raspberry Pi to control controllers and sensors
- GUI Monitor data and control the components
- Enclosure
- *Laser diode and photodetection to test FSO intensity*

Version:

16/02/2026

Added multiple threads

Added multiple DHT22 > SHT15

Added multiple BMP280

Added Data table

Changed temperature, RH to mean values

Experimental FSO in a foggy chamber

Proposal:

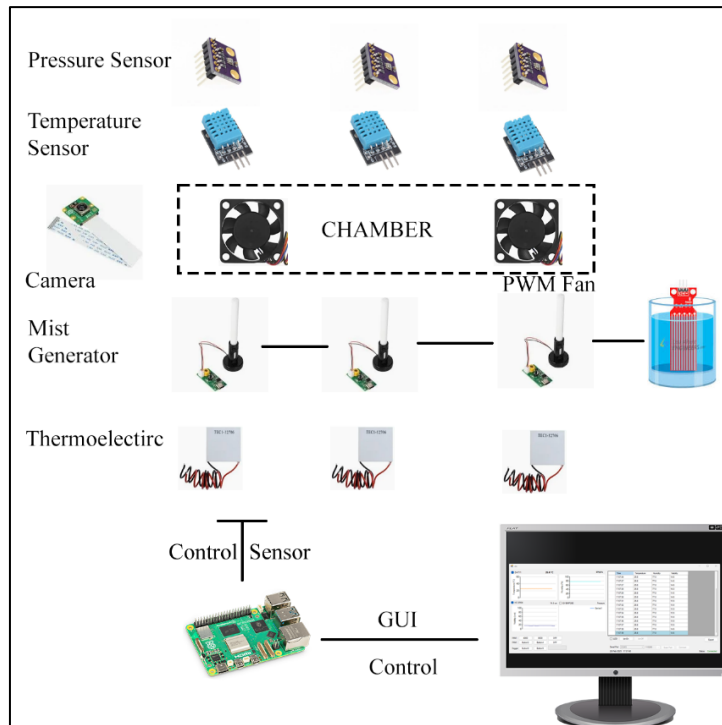
- Test the laser intensity of the RGB Laser diode in various environments. (Fog, Temperature, Wind speed, Distance)
- Validate the refractive index structure parameter (C_2)
- Test the Kim model from visibility (using laser intensity and camera)
- Test the pathloss model, **[Kim, Kruse, and Geometric path loss]**(using distance)*
- Simulate the test by Optism

System Components

Prototype: Raspberry Pi controlling the chamber Design

Raspberry Pi / Python / Sensors

- This can validate fog visibility by calculating the laser intensity and/or OpenCV capture LED array (Runway Visual Range Model).



This environmental chamber is designed to control fog and temperature for FSO experiments. A Raspberry Pi is the main controller for managing mist generators to produce fog, thermoelectric modules to control the temperature, and PWM fans to control wind speed.

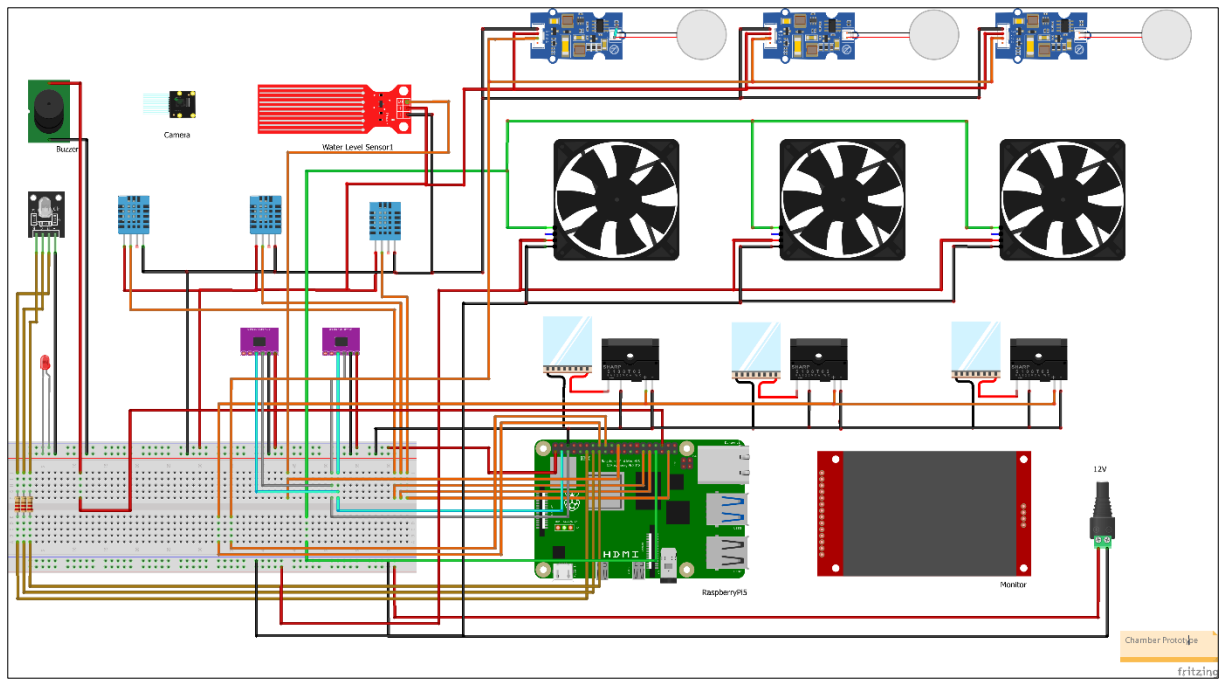
Sensors inside the chamber, such as pressure and temperature sensors, continuously send data to the Raspberry Pi. This data is displayed on a graphical user interface (GUI). Also, the GUI can be used to control the mist generators, thermoelectric units, and fans.

A camera module captures live video from inside the chamber. The video is displayed on the GUI and processed using computer vision (CV) algorithms to estimate visibility distance. Additionally, the GUI supports exporting data from sensors for further analysis.

A laser diode and photodetector measure fog. The system works as a transmissometer/photometer, comparing the initial laser intensity with the received laser intensity. This can be used to calculate fog visibility.

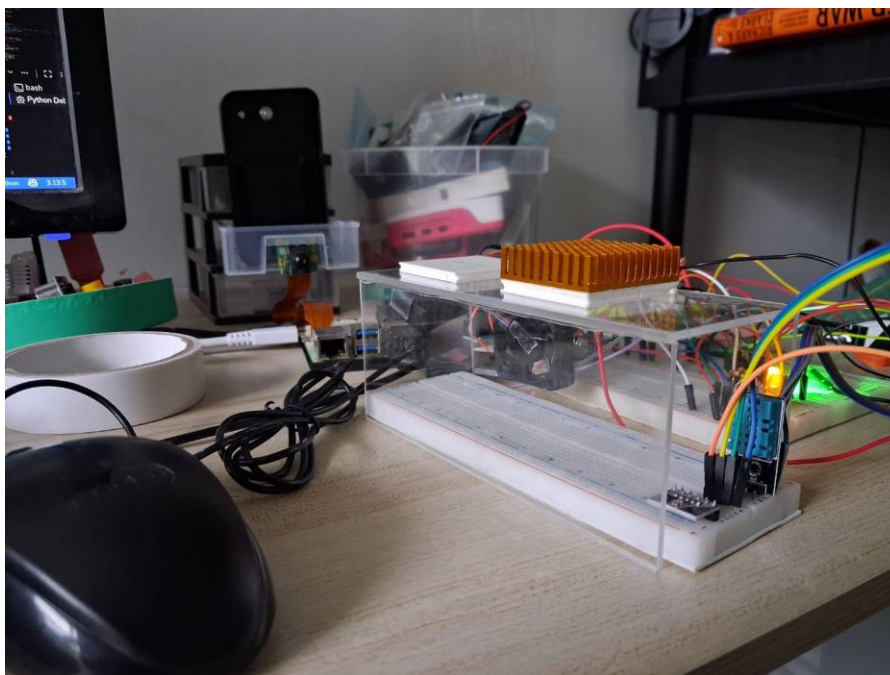
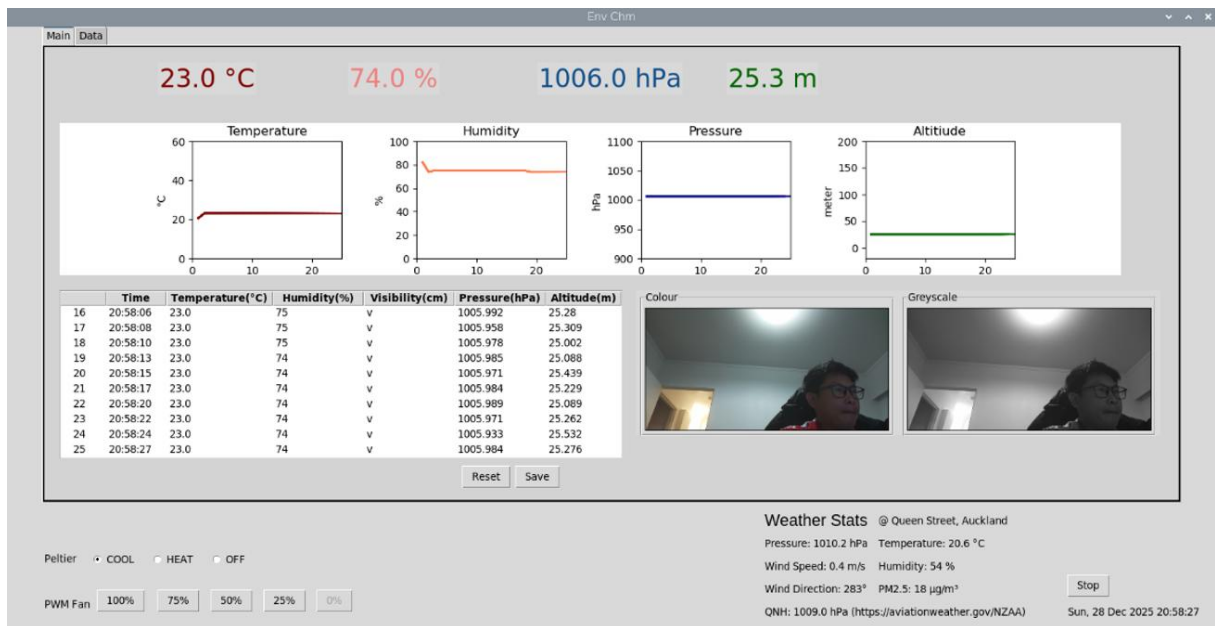
Overall, the chamber provides a controllable environment for FSO experiments under varying atmospheric conditions.

A Breadboard Wiring Diagram



Flow Chart
Block Diagram

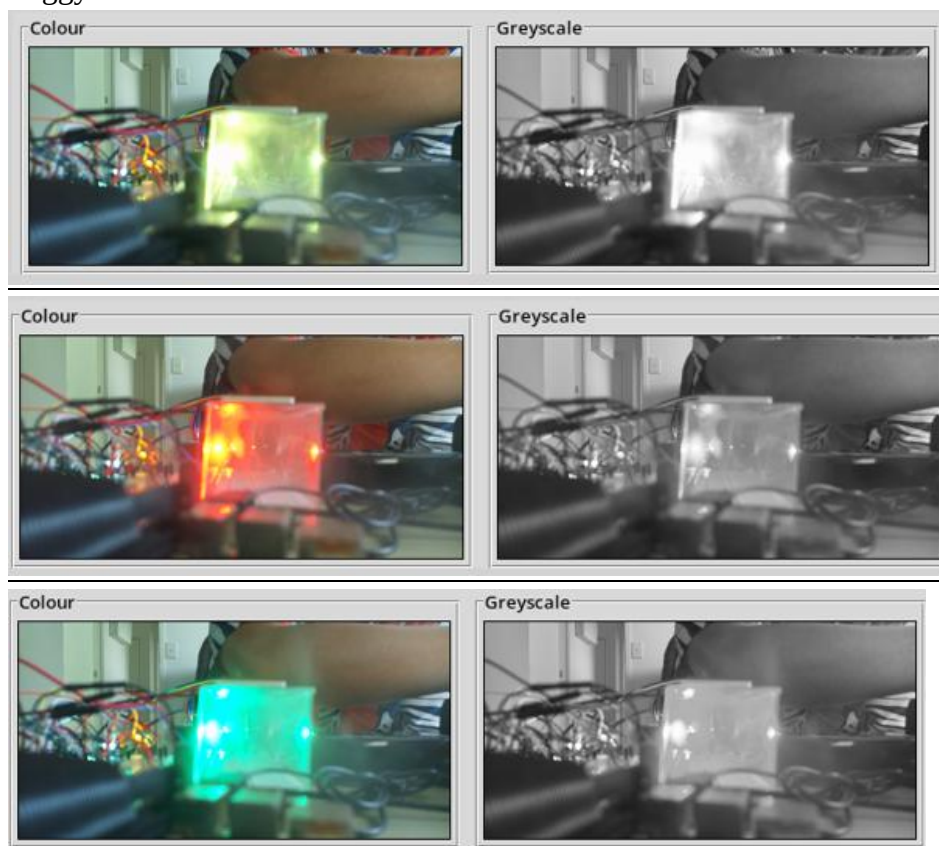
GUI



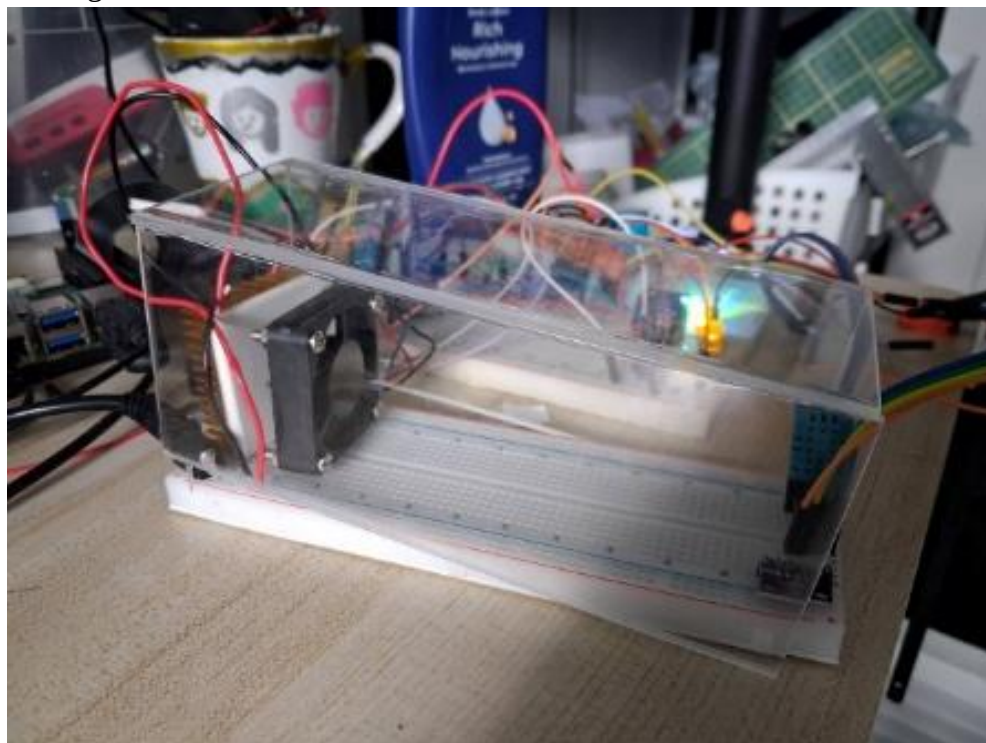
Camera

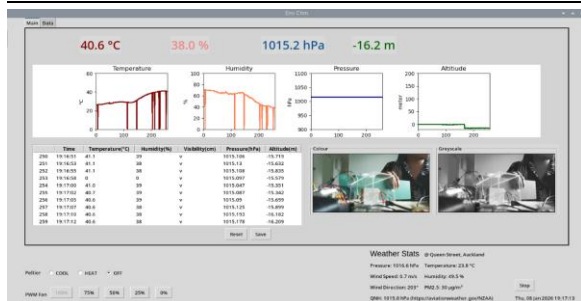


Foggy



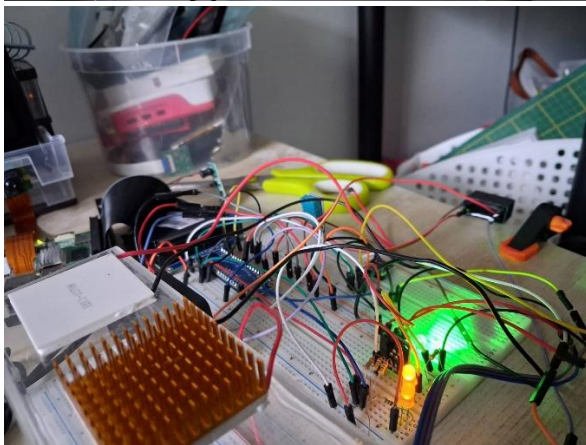
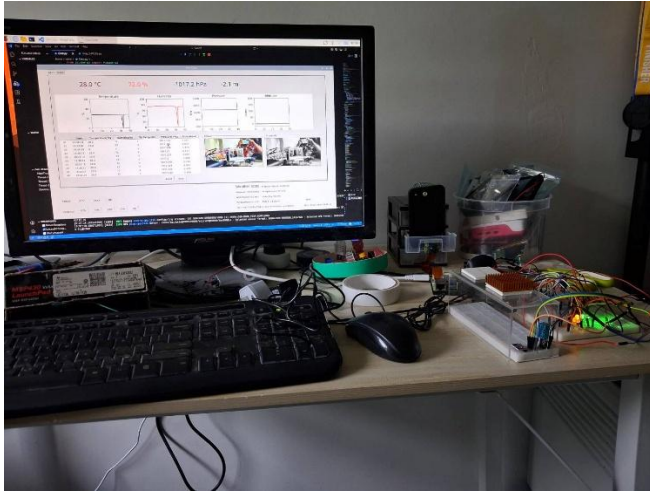
Testing





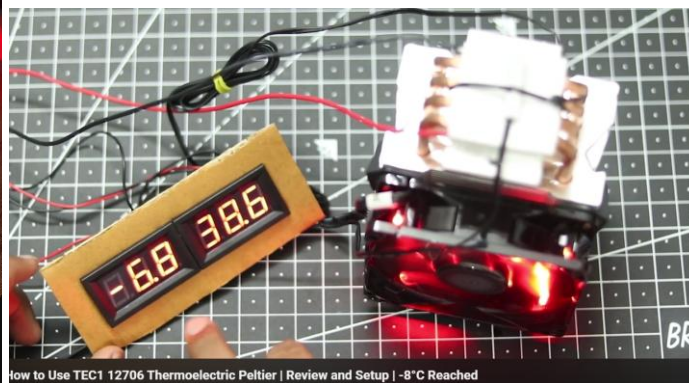
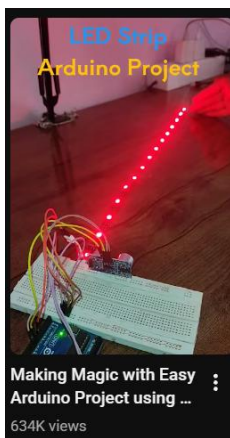
Room temperature: 28 C
Highest temperature: 41.1 C (+13.1 C)
Lowest temp: 19.4 C (-8.6)

Circuit

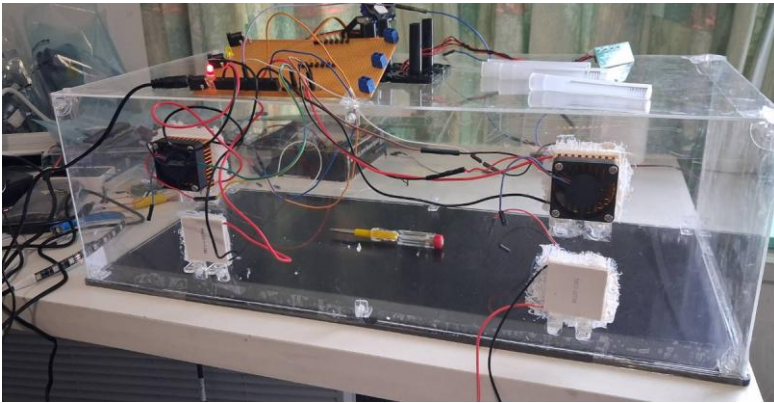


Note

Distance? Path Loss model? Camera? Vasibileter? LED stripe?



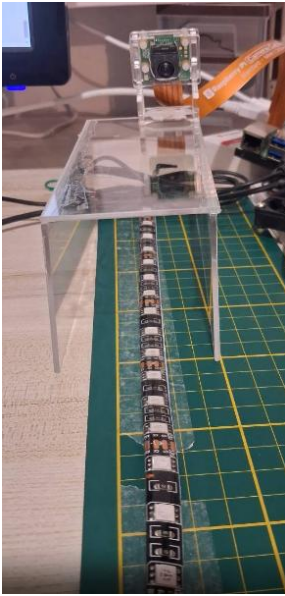
2nd Setup



01fc608e-ab77-4ecf
-abc9-89a77dfa6030

**Fogger /
Peltier X**

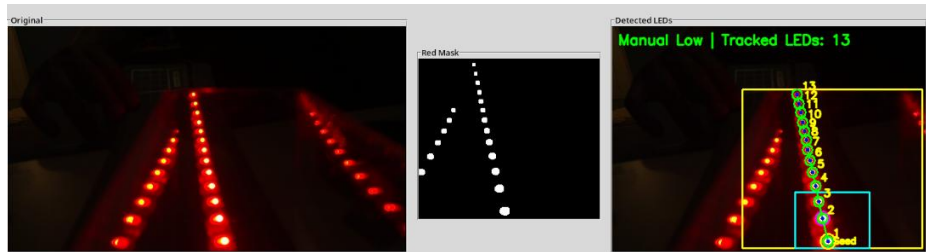
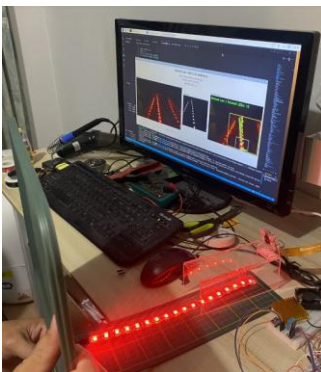
Camera + LED stripe Test

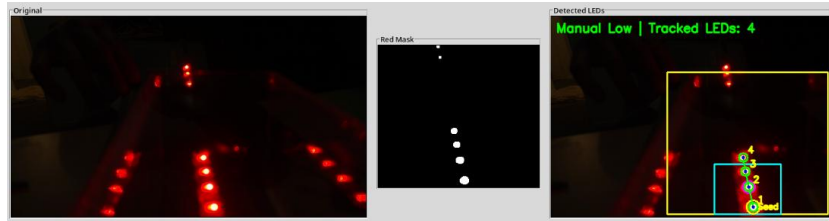
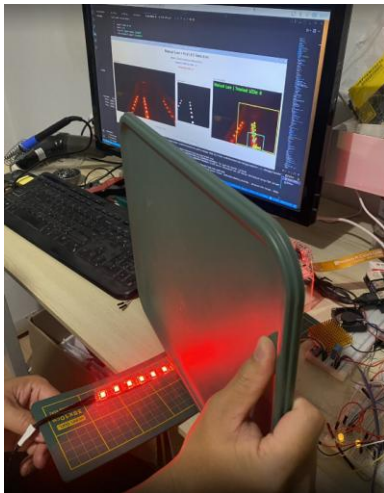


Configuration: Fix white balance (light adjustment), exposure gain (LED blooming)

Detection: Red LED change to black and white mask > detect the nearest visible LED in blue box
> track LED alignment > yellow box eliminate the light reflection

GUI





Remark

Need to tune the camera settings

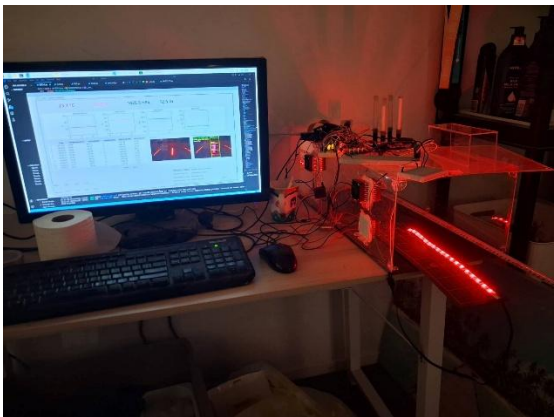
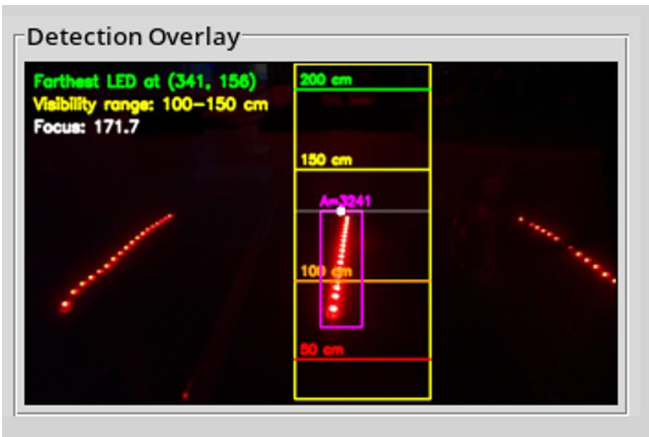
Camera setting: White balance/Focus/exposure gain

CV setting: colour shade / Line reference

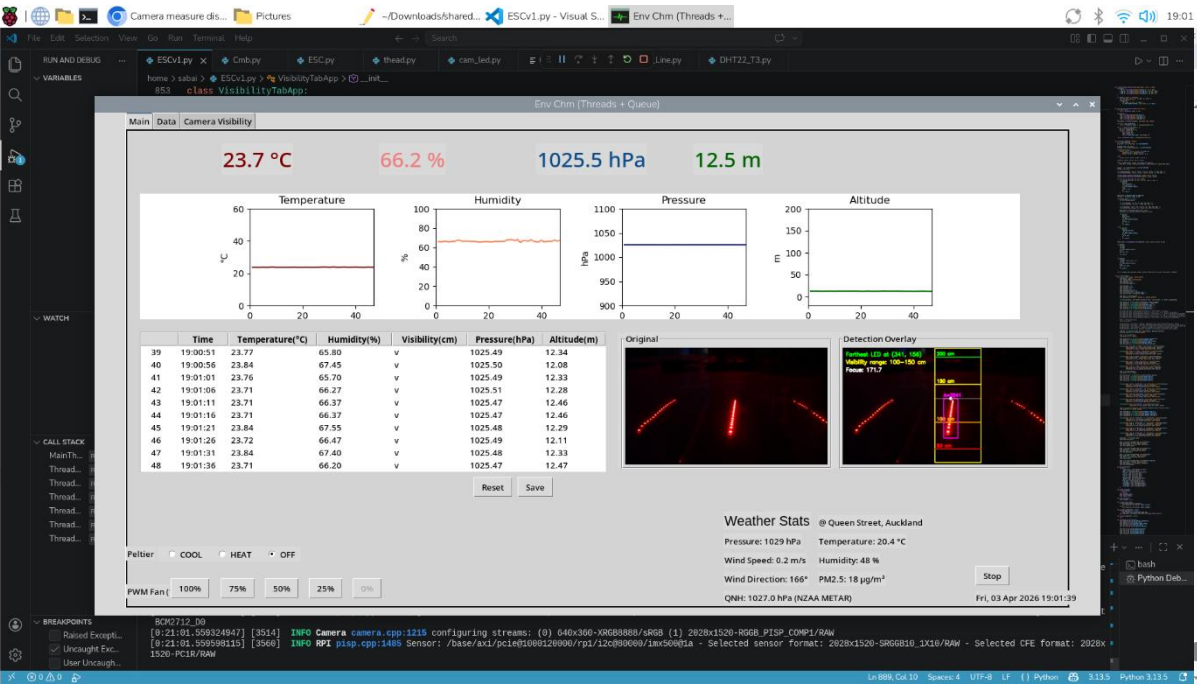
LiDAR test > Fail when fog occur

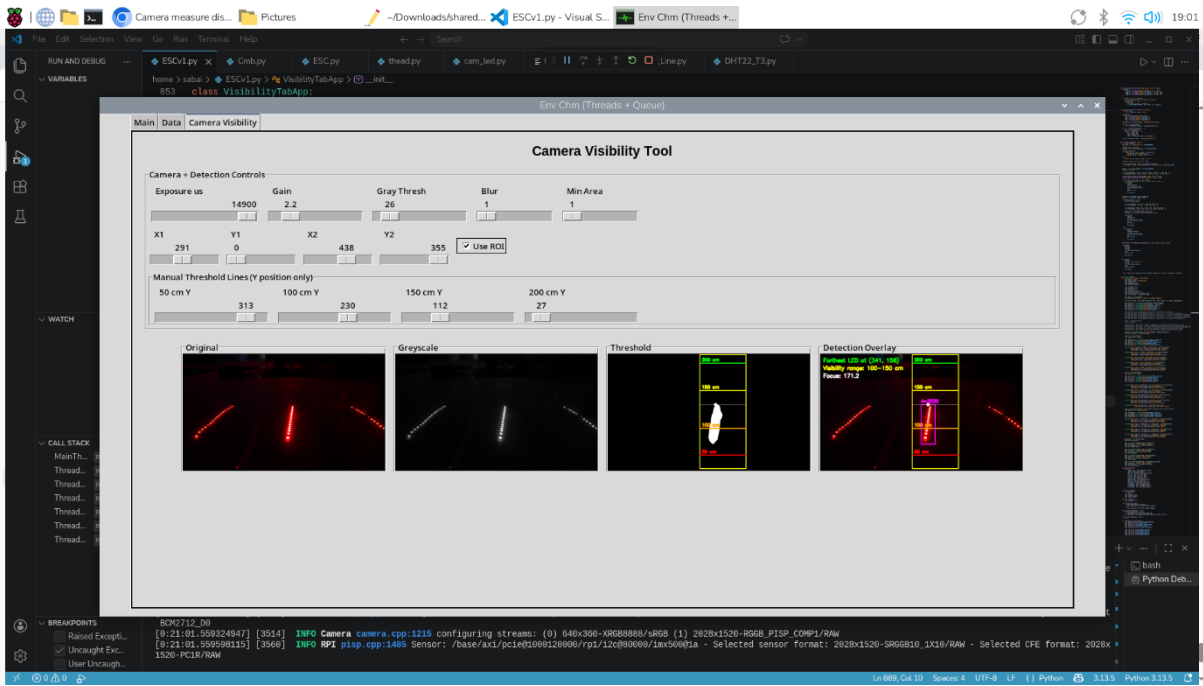
- Other studies use reference techniques such as /an object (Not suit/km) /bare eye (unreliable) /visibilimeter (cost)

LED strip



Pasted text.txt 0fef1f3d-d4b4-4736
-ae6b-2c97f57b7d6c





Experimental FSO in a foggy chamber

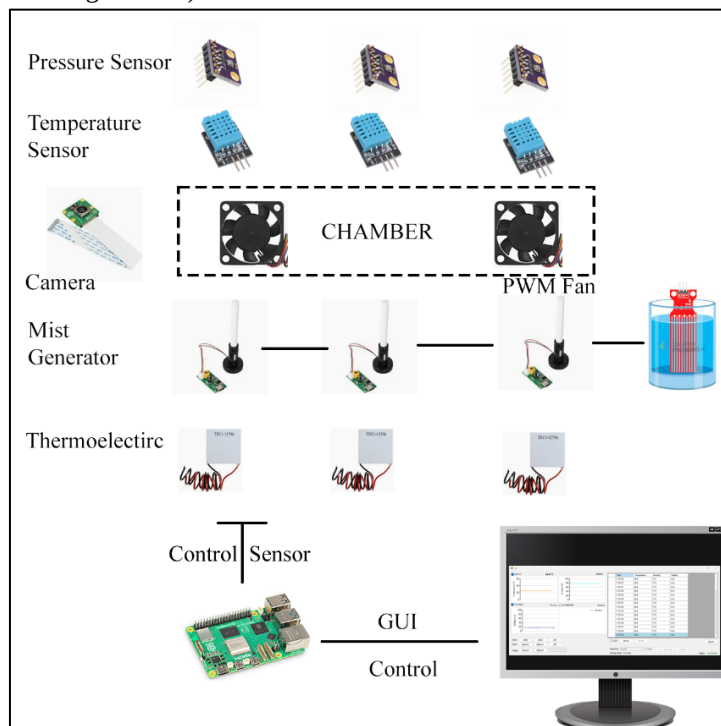
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- Validate the refractive index structure parameter (C_2)
- Test the Kim model from visibility (using laser intensity and camera)
- Test **the Malaga model** or other [Γ , ...](using distance)*
- Simulate the test by Optism

Prototype: Raspberry Pi controlling the chamber Design

Raspberry Pi / Python / Sensors

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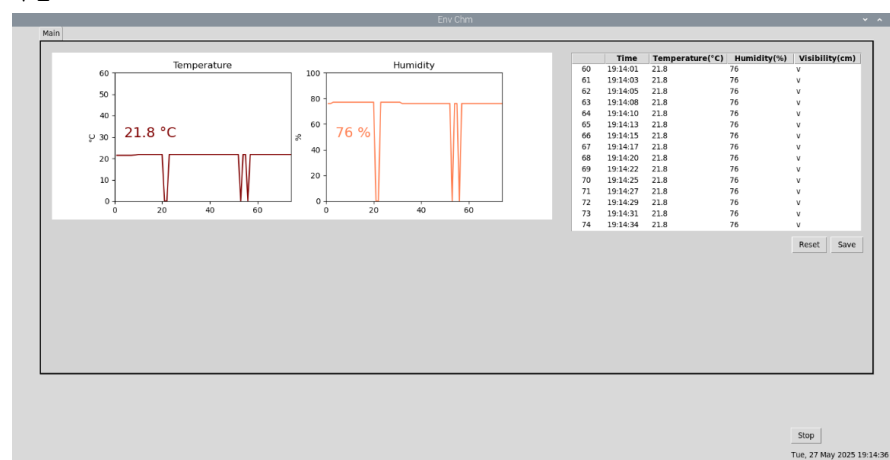
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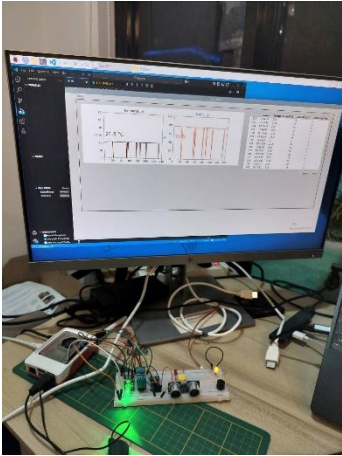
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On progression

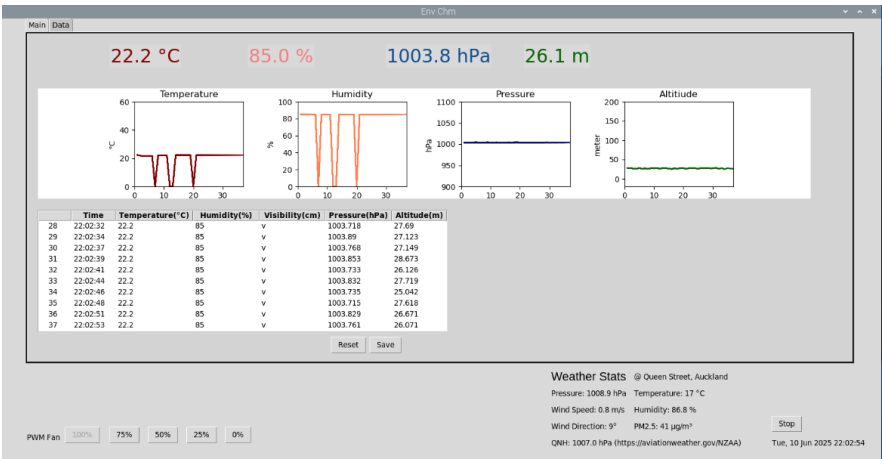
Sensors		GUI	Test
Temperature	SDH11 > SDH22	/	/
Humidity		/	/
Time		/	/
Pressure	BMP280 > BMP388	/	/
Laser+PD			
Camera	Narrow wide angle/		
Fog generator	Pressure sensors/ laser/ find some measurement/ spector meter		
PWM Fan		/	/
Cooler/Heater	TEC1-12706		
Enclosure	Acrylic		
MET Monitor	METSTAT.net	/	/

V1

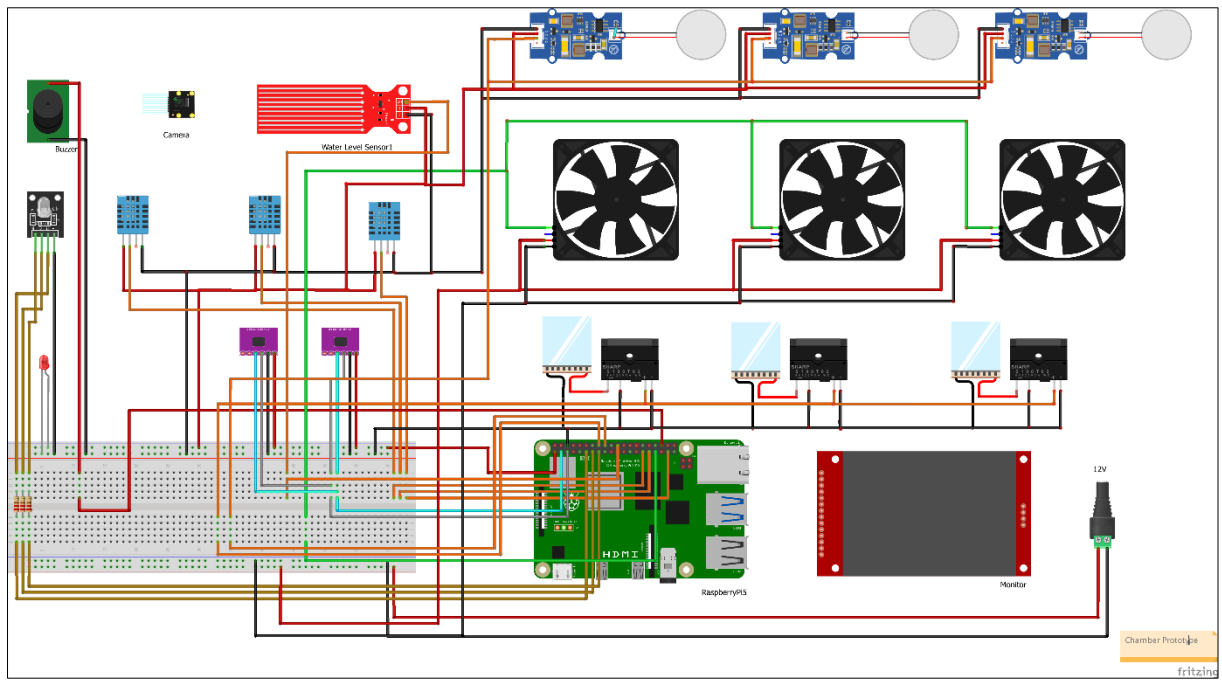




V2



TEST#3



Prototype2 design

1. Create GUI (Visual Studio) to control modules (ON/OFF/SPEED)
2. Connect PC with Arduino to control the modules
3. Test
4. Enclosure design / Still thinking

TEST#1

1. Test sensor modules
 - 1.1 DHT11 for temperature and humidity
 - 1.2 Ultrasonic for visibility
 - 1.3 Mist Generator to generate fog
 - 1.4 GY-BMP280 to measure pressure
 - 1.5 Water leveling sensor to cooperate with the mist generator
2. Send data to PC and show the data
 - 2.1 DHT11 control/show
 - 2.2 Ultrasonic control/show
 - 2.2.1 Maybe change to a camera and use OpenCV to measure visibility
 - 2.3 FAN control / Need to buy a new one due to this model cannot control airspeed.
 - 2.3.1 PWM Fan Purchase
 - 2.4 Fogger /still testing
 - 2.4.1 Must add a relay to turn ON/OFF the module (The module only have a push button)
 - 2.4.2 Relay purchase
 - 2.5 Pressure /still testing
 - 2.6 Water leveling sensor / Pass and need to combine code with main code

